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Effects of time-since-fire on the tussock dynamics of a dominant grass (*Themeda triandra*) in a temperate Australian grassland

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Abstract

Changes in tussock attributes and sward structure with time-since-fire were documented for the dominant tussock grass, *Themeda triandra*, at the Derrimut Grassland Reserve in southern Victoria, Australia. When the inter-fire interval exceeded 6 yr, the number of tillers per tussock and the total number of tussocks declined, and by 11 yr, few live tillers or tussocks remained in the sward. Below-ground biomass was also substantially lower at this time. With increasing time-since-fire, the canopy of live leaves was elevated high above the soil surface and dead leaves accumulated around and over the tussock bases. Productivity declined in long unburnt areas and by 11 yr without disturbance, the canopy "collapsed" upon itself, forming a thick layer of dead thatch over the soil surface. A single fire in an area previously unburnt for 12 yr did not immediately return the tussocks to a state more characteristic of a site with a 4 yr inter-fire interval burnt at the same time. Inter-fire intervals of $\leqslant 5$ yr would appear necessary to maintain the health and competitiveness of *Themeda triandra*. These findings have important implications for the maintenance of faunal habitat and the potential for weed invasion into remnant grasslands. © 1999 Elsevier Science Ltd. All rights reserved.

Keywords: Australia; Conservation management; Grassland decline; Inter-fire interval; Tillers

1. Introduction

It is widely recognised that fire and grazing regimes have influenced the diversity of temperate grassland and grassy woodland remnants in south-eastern Australia, as in other grassland regions of the world (e.g. Booysen and Tainton, 1984; Collins and Wallace, 1990; Belsky, 1992; Fensham, 1998). Most species-rich sites in south-eastern Australia have a history of frequent burning (i.e. 1–3 yr intervals) and/or minimal stock grazing (Stuwe and Parsons, 1977; Stuwe, 1986; Kirkpatrick et al., 1988; Benson, 1994; Lunt, 1995; Prober and Thiele, 1995; Morgan, 1998a). Species-poor remnants, on the other hand, are either grazed by domestic stock or only very infrequently disturbed by fire or grazing (Stuwe and Parsons, 1977; Tremont, 1994; Prober and Thiele, 1995). Species may be lost in the complete absence of sward disturbance because of competitive exclusion (Scarlett and Parsons, 1990; Gilfedder and Kirkpatrick, 1994; Morgan, 1997), sometimes irreversibly given that few native grassland forbs appear to form large, persistent soil seed banks (Gilfedder and Kirkpatrick, 1993; Lunt, 1995, 1997; Morgan, 1995, 1998a).

Fire and grazing regimes can also alter the structure of a grassland community and the population dynamics and vigour of its dominant tussock grasses. Numerous Australian studies have reported the effects of intensive disturbance regimes on dominant grasses, allowing Moore (1970) to summarize the general changes that occur in a variety of native temperate grasslands under heavy and continuous stock grazing. Principally, there is a shift from C4 to C3 grasses, from caespitose perennials to annuals, and from native to exotic species. Frequent, low mowing may also have similar effects on the survival of tall-growing tussock grasses (Chan, 1980).

The effects of the total exclusion of stock grazing and burning on dominant tussock grasses, however, are generally less well documented in temperate Australia. In grasslands dominated by *Themeda triandra* (Kangaroo Grass), Groves (1974) suggested that there was a curvilinear relationship between shoot biomass and time-since-fire; by 5 yr after fire, new shoot growth would balance death and decomposition of old leaves to maintain a constant biomass. McDougall (1989) noted that in an undisturbed sward (10 yr since fire or grazing) in southern Victoria, tussocks were fewer, had smaller

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basal areas, and were poorly rooted compared to *Themeda* tussocks in a frequently burnt area. Litter levels were also much greater in the undisturbed area and covered all the intertussock spaces. McDougall (1989) concluded that without fire or grazing, "long unburnt *Themeda* tussocks could be expected to disappear from the sward" and seedling recruitment would be minimal under the dead thatch. In the following 10 yr, *Themeda* health declined further as a result of continued fire exclusion, leading to the death of many tussocks (Lunt and Morgan, 1998).

At the Derrimut Grassland Reserve in southern Victoria, Australia, fire is considered a necessary conservation management tool to maintain native species diversity (Craigie and Stuwe, 1992), but it has been applied in an adhoc manner since the creation of the reserve in 1985. Some areas have been burnt three times over the last 12 yr (maximum inter-fire interval of 4 yr), whilst other areas have never been burnt. Lunt and Morgan (1999) reported on the floristic changes that occurred during the first 10 yr of reservation and found that most native species that had survived long-term grazing were also fairly resilient to the variable fire regimes imposed. Surprisingly, the least resilient species to infrequent burning and grazing exclusion appeared to be the dominant tussock grass, *Themeda triandra*. This finding has important implications for the threatened fauna that rely on the grassland for habitat (e.g. the Striped Legless Lizard, *Delma impar*), as well as the susceptibility of the grassland community to invasion by exotic perennial grasses (Lunt and Morgan, submitted). In this paper, we investigate how time-since-fire affects the survival, vigour, structure and productivity of *Themeda triandra* tussocks, and the subsequent ability of tussocks to recover from fire.

2. Methods

2.1. Study site

The 154 ha Derrimut Grassland Reserve is 15 km south-west of Melbourne, Victoria (37°48′30″S, 144°47′40″E). Mean annual rainfall is 568 mm. The geology is Early Pleistocene olivine basalt (Douglas, 1982). The reserve is dominated by a *Themeda triandra* grassland in well-drained areas (Lunt, 1990a) and is a remnant of the Keilor Plains grassland described by Willis (1964). This grassland type is on the verge of extinction (McDougall and Kirkpatrick, 1994). Soils in grassland areas range from silty-clay to clay-loam with pH 7.0 (Lunt, 1990a).

The property was grazed by sheep until the 1960s and by cattle from then until 1985, when grazing was removed from the entire site after management changed to the state conservation agency for the purposes of grassland conservation.

Early recommendations for management of the reserve suggested that fire at 2–5 yr intervals were necessary to maintain botanical values (Stuwe, 1986; Lunt, 1988). However, owing to perceived negative impacts of frequent burning on endangered fauna (Coulson, 1990), a management plan recommended that the reserve be divided into 10 zones, two of which were to be burnt every year, giving a 5 yr fire interval in each zone (Craigie and Stuwe, 1992). In practice, the management plan has not been consistently implemented. Ten fires occurred between 1986 and 1997 but all had different boundaries and burnt different areas. The number of fires experienced in each zone ranged from 0 to 3 and the maximum inter-fire interval varied from 4 to over 12 yr (in areas which were never burnt).

2.2. Study species

Themeda triandra (syn. *T. australis*) is a perennial C4 tussock grass that dominates many grassy woodlands and tropical savannahs in both Australia and Africa. In southern Victoria, *T. triandra* is the dominant species of natural grassland communities on volcanic soils (Patton, 1935; Willis, 1964; McDougall et al., 1994). It grows primarily from October to January in southern Victoria and is most productive in the first 3–4 yr after fire (McDougall, 1989), after which time productivity declines (Groves, 1974; McDougall, 1989). It has no means of extensive vegetative spread. Tussocks are generally fire-resistant, tillers resprouting after summer fires from protected basal apices, even when burnt annually (Morgan, in prep). Aerial tillers form when swards remain undisturbed and these are easily damaged by trampling (McDougall, 1989). The ability of tussocks and tillers to recover from fire after a long period of fire exclusion is virtually unknown, although in a companion study, Lunt and Morgan (1998) found that post-fire recovery of *Themeda* in areas unburnt for 17 yr lagged behind that of a grassland burnt more frequently. Longevity of tussocks is unknown in Victorian grasslands; Morgan (1998b) observed little mortality and turnover of *Themeda* tussocks over a 3 yr period in a biennially burnt grassland. Seed set occurs primarily in mid-summer (January; McDougall, 1989) but germination does not occur until spring (October) when soil moisture is high and soil temperatures are $>20°C$ (Hagon, 1976). A thick litter layer of *Themeda* inhibits germination; seedlings are generally only observed after fire when the insulating vegetation cover and litter layer has been removed (Lock and Milburn, 1971; McDougall, 1989), or when grazed.

2.3. Changes in Themeda tussocks with time-since-fire

A chronosequence study was used to assess changes in *T. triandra* tussock structure with the time since last

fire. Areas of the reserve that originally supported structurally similar species-poor *Themeda* grassland at the time of reservation in 1985 (Lunt, 1990a), and that subsequently differed in their fire histories, were chosen for comparative study.

The design of this study clearly represents a case of pseudoreplication (Hurlbert, 1984); whilst there was replication of plots within each of the fire history treatments (each >10 ha in size), there was no replication of the treatments (fire histories) themselves. There was no opportunity to achieve a statistically valid design to experimentally test for the effects of time-since-fire on *T. triandra* but rather, only the chance to study a single grassland that had, by fate of the management over the last decade, differed in fire history (i.e. the "unplanned or natural experiment" of Diamond (1986)). Sharp boundaries between the different fire zones were clearly discernible on the ground and hence, differences in tussock attributes between treatments were unlikely to be due to soil factors. Such a design places restrictions on the application of inferential statistics, but need not prevent ecological interpretation of the data and their implications for management of the Derrimut Grassland Reserve.

In areas that had last been burnt 1, 3, 5 yr ago (number of years are expressed as integers) or had remained unburnt (11 yr since exclusion of disturbance), five 0.25 m^2 quadrats were randomly located in visually homogeneous vegetation over an area of ca. 1 ha and the number of live tussocks, live rooted tillers and inflorescences of *T. triandra* were counted in December 1996. Basal circumference of tussocks was also measured. Most tussocks usually had an irregular shape that prevented accurate basal diameter estimates.

Total root biomass was determined in November 1997 in areas burnt 2, 4 and 12 yr ago. At 2 m intervals along a single transect in visually homogeneous vegetation in each fire zone, and nearest the closest, large (>10 cm diameter) tussock, one soil core (2 cm diameter × 10 cm depth) was taken within 3 cm of the edge of the tussock. A total of 18 cores were collected per fire zone. The soil samples were then sieved and rinsed under a jet of water before drying the root matter at 80°C for 72 h. It was assumed that most of the roots recovered were from *T. triandra*, an assumption deemed reasonable given that few other species were observed in the areas sampled.

Changes in vegetation structure in areas last burnt 0.2, 1, 3, >6 and 11 yr ago were documented using point quadrats (Kent and Coker, 1992). A frame consisting of 10 vertically held 4 mm diameter pins, each 10 cm apart, was randomly placed at 12 locations within each fire zone ($n = 120$ points per fire zone) and all hits of both live and dead *T. triandra*, and their height of contact (in 3 cm intervals), were recorded.

Seasonal changes in above-ground biomass in zones burnt 1, 3 and 11 yr ago were followed for 18 months, from August 1995 to February 1997. At each site, 12 random 0.25 m^2 quadrats were selected at 4–6 week intervals for six months, and then again 12 months later, and all vegetation was cut to ground level, dried at 80°C for 48 h and weighed. It was assumed that most (>90%) of the biomass was that of *T. triandra*, an assumption likely to be valid given that (1) few other species occurred in the areas sampled and (2) the high representation of *T. triandra* in the biomass samples hand sorted by McDougall (1989) in a similar grassland nearby.

The comparative ability of *Themeda* tussocks to recover from fire was assessed by burning a 1 ha area, unburnt for 12 yr, and an equivalent area most recently burnt 4 yr previously. Both areas were burnt in November 1997. In April 1998, the total number of rooted tillers and tussocks, tussock basal circumference and visual percent cover were recorded in five randomly placed 0.25 m^2 quadrats at each site.

2.4. Statistical analyses

The number of tussocks per 0.25 m^2, number of live tillers, average number of tillers per tussock (square root transformed), number of inflorescences (square root transformed), average number of inflorescences per tussock and mean basal circumference were compared between fire zones that differed in their time-since-fire by one-way analysis of variance (ANOVA) with Tukey's post-hoc comparisons (Zar, 1984). Homogeneity of variance assumptions were tested using Cochrane's Test (Winer et al., 1991). A mixed-model nested ANOVA was used to test whether basal circumference was affected by site history, with quadrats nested within fire zones (Sokal and Rohlf, 1981). The *G*-test of independence was used to test whether basal circumference, when grouped into size classes, was associated with fire history (Sokal and Rohlf, 1981). Root biomass differences with fire history were compared using a one-way ANOVA on log transformed data with Tukey's post-hoc comparisons.

Attributes of post-fire recovery of areas previously unburnt for 4 and 12 yr (i.e. number of tussocks per 0.25 m^2, number of tillers, average number of tillers per tussock, percent cover) were compared using *t*-tests (Sokal and Rohlf, 1981). Homogeneity of variance assumptions were checked using the F_{max}-test (Sokal and Rohlf, 1981) and satisfied in all cases. A contingency table was used to test whether basal circumference, when grouped into size classes, was associated with fire history. Expected values were calculated based on the null hypothesis of independence and tested using the chi-square statistic (Zar, 1984). Standardised residuals were examined to determine the pattern of association between the variables.

3. Results

All of the above-ground tussock attributes, except basal circumference, significantly declined with increasing time-since-fire (Table 1). There were no significant differences in tussock attributes between areas burnt 1 and 3 yr previously. Rather, the effects were significant when grasslands had remained unburnt for $\geqslant 5$ yr.

The total number of tussocks per 0.25 m^2 was lowest in the long unburnt area (11 yr; $F=7.10$, $p=0.003$) but did not differ significantly between areas burnt more recently ($\leqslant 5$ yr). Total number of tillers and mean number of tillers per tussock were highest in 1 and 3 yr unburnt areas, intermediate, but significantly lower, in the 5 yr unburnt area, and least in the 11 yr unburnt area ($F=35.32$, $p<0.0001$ and $F=10.54$, $p<0.001$ respectively). Inflorescence production (total and mean number per tussock) declined with time-since-fire in a similar way ($F=47.91$, $p<0.0001$ and $F=16.82$, $p<0.001$ respectively). Tussock circumference did not differ within ($F=0.926$, $p=0.540$) or between ($F=1.464$, $p=0.224$) any of the fire zones, nor was the size class distribution of tussocks significantly associated with time-since-fire ($\chi^2=12.07$, $n=12$, $p>0.05$). Thus, all tussock size classes were represented in all fire zones in similar ratios.

Mean root biomass ($\pm$1SE) in the 12 yr unburnt area (0.174 ± 0.019 $g/100$ cm^3; $F=26.19$, $p<0.001$) was less than 45% of that in the areas burnt two (0.432 ± 0.016 g/100 cm^3) and four (0.384 ± 0.023 g/100 cm^3) years previously, which did not differ from one another ($p>0.05$).

While the total number of contacts with live *Themeda* did not increase greatly in the years 1 to >6 after burning in the point quadrat study (Fig. 1), the spatial distribution of live leaves in the canopy changed dramatically (Fig. 2). Immediately after fire, *Themeda* resprouted rapidly from the base and live leaves were mostly concentrated near the ground level ($\leqslant 6$ cm) during year 1. In years 2 to >6, the canopy of live leaves elevated higher in each year so that by >6 yr, most of the live canopy was >18 cm above ground level. During the same period, live leaves disappeared completely from tussock bases (<3 cm) and dead vegetation accumulated from the 'bottom up' (Fig. 2). At >6 yr after fire, 73% of all *Themeda* leaves in the canopy were dead. After 11 yr without disturbance, the grassland appeared to collapse upon itself (Fig. 2). Dead vegetation remained on the soil surface (in a thick layer to 12 cm deep) whilst almost no live vegetation existed at any layer in the canopy.



Fig. 1. Changes in the total number of live and dead leaves of *Themeda triandra* (per 100 point quadrats) with increasing time-since-fire.

Over the 18 months in which above-ground biomass was followed, areas unburnt for 1 and 3 yr at the start of observations increased their total biomass by 167% and 152%, respectively (Fig. 3), within the context of large seasonal variation. During the same time, the total biomass of the area unburnt for 11 yr declined to 58% of its original amount. The decline was most dramatic between February 1997 and February 1998 when biomass fell from 707 g m^{-2} to 462 g m^{-2}, despite large increases in biomass at the other two sites. This decline corresponded with the grassland collapse observed in areas which had remained unburnt for >12 yr.

Time-since-fire affected the subsequent ability of *Themeda* tussocks to recover from fire (Table 2). The canopy of the long unburnt area (12 yr since fire) regrew more slowly (7% versus 35%), had fewer tussocks per 0.25 m^2, had fewer tillers and fewer tillers per tussock than the area that had a shorter inter-fire interval (4 yr since fire; Table 2). Whilst there was no significant

Table 1
Tussock attributes of *Themeda triandra* (mean $\pm$1SE, $n=5$) with time since last fire. Significant differences are indicated by different row superscripts (lower case: $p<0.01$; upper case: $p<0.001$)

Tussock attribute	Time since last fire (years)			
	1	3	5	11
No. tussocks/0.25 m^2	19.2 ± 1.6^a	21.0 ± 1.9^a	21.2 ± 2.5^a	10.8 ± 1.0^B
No. tillers/0.25 m^2	383.4 ± 28.9^a	374.2 ± 20.2^a	230.6 ± 23.7^B	103.0 ± 14.2^C
No. tillers/tussock	20.9 ± 3.0^a	18.2 ± 1.3^a	11.0 ± 0.5^B	9.7 ± 1.4^B
Basal circumference (cm)	13.5 ± 1.5^a	11.3 ± 1.1^a	10.2 ± 1.3^a	10.5 ± 0.3^a
No. inflorescences/0.25 m^2	79.8 ± 11.8^a	90.6 ± 4.5^a	19.0 ± 4.1^B	8.2 ± 2.6^B
No. inflorescences/tussock	4.2 ± 0.7^a	4.5 ± 0.7^a	0.9 ± 0.2^B	0.8 ± 0.3^B

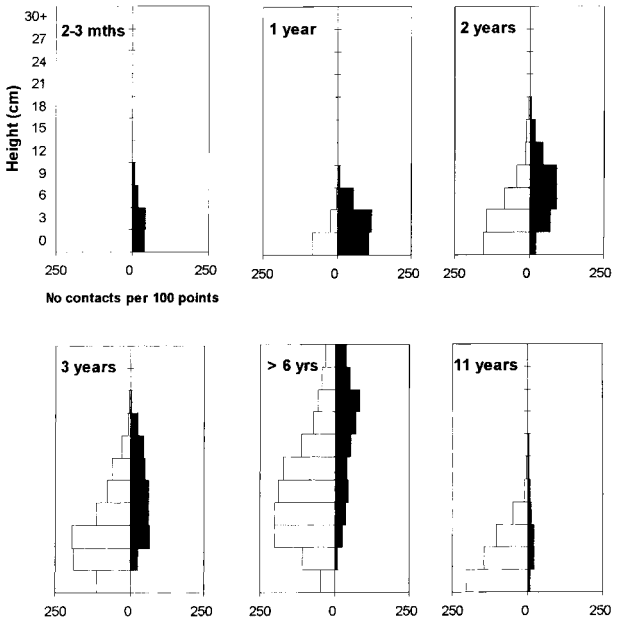


Fig. 2. Changes in the height distribution of live (■) and dead leaves (□) of *Themeda triandra* (per 100 point quadrats) with increasing time-since-fire.

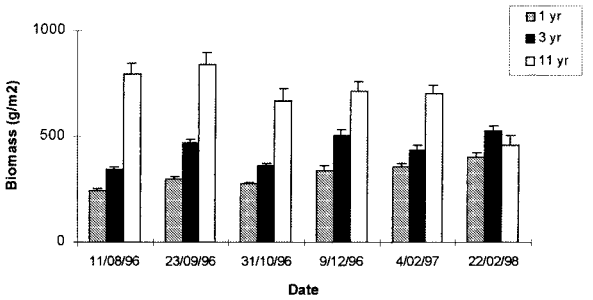


Fig. 3. Changes in total above-ground biomass (mean ±1SE) over an 18 month period in *Themeda triandra* grasslands initially unburnt for 1, 3 and 11 yr.

difference in the mean basal size of tussocks at each site (Table 2), the size class distribution of regenerating tussocks was associated with time-since-fire (Table 3). Tussock regeneration at the long unburnt site consisted of significantly fewer large tussocks (>25 cm basal circumference) and significantly more intermediate-sized tussocks (5–15 cm) than expected using a null hypothesis of independence, whilst at the 4 yr burn site, large tussocks (>25 cm) were more common than expected.

4. Discussion

It has long been recognised that frequent biomass removal by burning is necessary to maintain native plant diversity in ungrazed, species-rich grassland remnants on the western volcanic plains in southern Victoria (Stuwe and Parsons, 1977; Scarlett and Parsons, 1982; Stuwe, 1986). Where inter-fire intervals exceed 5 yr, even on only rare occasions, competitive exclusion by the dominant tussock grass (*T. triandra*) has usually been considered to reduce native plant diversity (Stuwe and Parsons, 1977; Scarlett and Parsons, 1990). This decline may be irreversible given that the soil seed bank type possessed by many native species is a transient one (Lunt, 1995, 1996; Morgan, 1995, 1998a). This study suggests that frequent biomass removal is also necessary to maintain the health and vigour of the dominant grass itself at one of the few reserves dedicated to the conservation of this threatened community (McDougall and Kirkpatrick, 1994).

The effect of 'time-since-fire' on *Themeda* was the specific parameter of interest in this study, but it is confounded with another fire regime factor, namely, the fire frequency (Morrison et al., 1995). Some areas experienced no fires in the 12 yr since reservation, others only one fire, and yet other areas experienced three fires in the same period. Although the effects of fire frequency on *Themeda* tussocks should not be discounted, given that all fire zones were generally similar at the time of reservation and that the number of fires experienced at all sites was low, the parameter of time-since-fire would appear to be a valid one to compare between fire zones.

When time-since-fire exceeded 6 yr, the total number of tillers, number of tillers per tussock and number of inflorescences per tussock all declined substantially. Whilst the total number of live leaves in the canopy was relatively constant between yr 2 and 6 after fire, the canopy was raised high above the soil surface with a consequent accumulation of dead leaves around the tussock base. In the prolonged absence of fire (by 11 yr),

Table 2
Tussock attributes (mean ±1SE, $n = 5$) at approximately six months after burning in *Themeda triandra* areas previously unburnt for 4 and 12 yr

Tussock attribute	Years since previous fire		Significance p
	4	12	
% *Themeda* cover	35 ± 2	7 ± 3	<0.0001
No. tussocks/0.25 m^2	28.2 ± 2.7	7.3 ± 3.2	<0.001
No. tillers/0.25 m^2	285 ± 16.7	59.2 ± 14.7	<0.0001
No. tillers/tussock	10.7 ± 1.5	6.3 ± 1.1	0.052
Basal circumference (cm)	9.2 ± 1.5	6.4 ± 1.2	>0.05

Table 3
Post-fire tussock size class distribution in grassland previously unburnt for 4 and 12 yr. Values are observed and expected (in parentheses) frequencies based on the null hypothesis of independence

Tussock circumference class (cm)	Years since previous fire	
	4	12
< 5	82 (79)	23 (26)
5–15	29 (35)	18 (12)
15–25	12 (13)	5 (4)
> 25	18 (14)	0 (4)
Total	141	46

$\chi^2 = 11.26$, $n = 3$, $p < 0.05$

tussock degeneration becomes substantial; few live tussocks survive in the grassland and the canopy collapses on itself. At 11 yr after fire, the grassland consists mainly of a dead, slowly decomposing thatch, with little live above-ground material and a markedly declining productivity. The decline was mirrored below-ground where root biomass was also substantially lower than in areas burnt more recently. McDougall (1989) reported the beginnings of tussock decline in a *T. triandra* grassland reserve nearby after 10 yr of fire exclusion and we have observed similar processes in other *Themeda* grasslands in western Victoria, Gippsland and the Australian Capital Territory. Hence, the phenomenon is likely to have some generality in *Themeda*-dominated grasslands in south-eastern Australia. The rate of tussock decline will undoubtedly vary between sites, and depend chiefly on site productivity. Interestingly, similar responses to fire and grazing exclusion have also been reported for *T. triandra* elsewhere and this suggests that it is a feature of the species itself, rather than a local effect restricted to southern Australia. Mott and Andrew (1985), for instance, found that *T. triandra* tussocks senesced rapidly after 3 yr without fire in a tropical savanna in north-west Australia whilst Belsky (1992) has reported that *T. triandra* can decline in less than 5 yr in the absence of disturbance in a tropical grassland in East Africa.

The causes of *Themeda* tussock decline with increasing time-since-fire were not studied here and are poorly understood. It is suspected that shading (i.e. light quantity) at the soil surface, due to the increase in above-ground biomass with time-since-fire, may be a primary cause of tussock decline, as has been reported in other productive grasslands (e.g. Hulbert, 1969; Old, 1969; Knapp and Seastedt, 1986). Photosynthetically active radiation (PAR) at the ground surface can be < 5% of PAR available above the canopy in *Themeda* grasslands unburnt for > 4 yr (Morgan, 1997), due to the reflection and absorption of energy by the standing plant material (Knapp, 1984). Hence, the need to elevate the photosynthetic surface high above the ground increases with time-since-fire as biomass accumulates from the ground up. Some *Themeda* tillers have the ability to elevate well above the soil surface (described as 'aerial tillering' by McDougall (1989)) to reach the light, but many tillers do not successfully compete for light in this way, leading to the loss of live tillers and the resources that are captured by the tussock. A point is reached (by 11 yr after fire) when aerial tillers can no longer be elevated higher on their weak stems and the canopy collapses. The dead material forms a thick thatch that effectively mulches the soil and prevents seedling regeneration (Lock and Milburn, 1971; McDougall, 1989) and new tiller initiation.

Other factors that potentially contribute to *Themeda* tussock decline with increasing time-since-fire also reflect the effects that standing dead and litter material have on microclimate at the tussock bases. Two factors appear to warrant further consideration, namely: (i) the effects of red:far red light (i.e. light quality) (Deregibus et al., 1985) and, (ii) below- and above-ground temperatures (Old, 1969; Knapp, 1984). Both factors have been implicated in the abilities of other tussock grasses to initiate new tillers, with modifications in light quality by a dense grassland canopy being particularly detrimental to tiller populations of some other C4 grasses (Deregibus et al., 1985).

After a long fire-free period (12 yr), a single fire did not, in the short-term, return the grassland's tussock structure to a state more characteristic of a site with a shorter inter-fire interval (i.e. 4 yr). There were fewer tillers and tussocks per 0.25 m^2 in the long unburnt site six months after fire relative to the shorter fire-interval site and, in general, the tussock structure resembled that in the pre-fire grassland. The rate of *Themeda* canopy recovery was also slower in the long unburnt area. This has also been observed in a nearby grassland when fire was reintroduced after 17 yr (Lunt and Morgan, 1998). Slow vegetative recovery probably results because pre-fire tiller health was poor, with many aerial tillers being killed by fire, and hence, the subsequent regrowth is from the relatively few, newly initiated tillers found at the soil surface. Regrowth from tillers in the more frequently burnt areas, however, is more likely to be from the recovery of existing tillers, protected by the densely crowded meristems in the basal tuft (Morgan, 1998b). Whether *Themeda* tussock recovery in the long unburnt area continues in the subsequent post-fire yr, enabling the grassland to more closely resemble the frequently burnt area, is unknown. At the nearby Laverton North Grassland Reserve, however, areas burnt after 17 yr of fire exclusion appear to retain their differences (i.e. low tussock density, less canopy cover) at 3 yr after burning compared to adjacent areas burnt more frequently (Morgan, pers. observation).

The decline in *Themeda* health with long time-since-fire intervals has two major implications for conservation management at Derrimut. Firstly, changes in

grassland structure may affect the habitat suitability of the site for the endangered *Delma impar* (Striped Legless Lizard), a species largely dependent on the western volcanic plains grassland community. The animal's habitat requirements have not been conclusively determined (e.g. Coulson, 1990; Moro, 1990) but are presumed to revolve around the dense *Themeda* cover that develops in the yr after fire. Further research is clearly needed to identify the structural requirements most favoured by *D. impar*, perhaps using a technique similar to the point quadrat method used here to quantify grassland structure. However, it is apparent from this study that excluding fire for long periods is likely to cause sward deterioration and hence, potential habitat loss for this species in the long-term.

Secondly, *Themeda* sward health may play a role in resisting invasion into the grassland by the aggressive alien perennial grass *Nassella neesiana* (Lunt and Morgan, submitted). *Nassella neesiana* is perhaps the most serious environmental weed in remnant native grasslands in southern Victoria (McDougall and Kirkpatrick, 1994; McLaren et al., 1998) and poses a major threat to the conservation of this endangered ecosystem. When *Themeda* swards are productive and dense ($>50\%$ live cover) and tussock densities are high, invasion appears to be inhibited relative to areas where the *Themeda* sward cover is low and/or unhealthy (Lunt and Morgan, submitted). Leaving grassland areas unburnt for long periods is likely to increase the potential for, and rate of, invasion by *N. neesiana* because of the decline in vigour and competitive ability of *Themeda*. Maintenance of a healthy sward of *Themeda* may prove to be an extremely cost-effective method of slowing (but not preventing) invasion by *N. neesiana* at Derrimut.

In summary, the productive, ungrazed grassland at Derrimut requires frequent biomass removal via fire if the structural dominant (*T. triandra*) is to remain in a vigourous state. Short inter-fire intervals (i.e. <5 yr intervals) will allow *Themeda* to maintain its competitiveness and the habitat values it provides for endangered vertebrates in the long-term. Burning will also benefit an inter-tussock native flora that can be rapidly eliminated from grasslands due to severe competition from *Themeda* in the absence of disturbance. Frequent ecological burning is therefore likely to be a far more cost-effective conservation management option at Derrimut than the current practice of intermittent, and largely unplanned, burning. When the time between ecological burns exceeds 5–6 yr, however, substantial costs are likely to be incurred to (a) restore degenerate *Themeda* swards again to a healthy condition and, (b) control aggressive perennial weeds that flourish with long inter-fire intervals. Such activities are non-sustainable in the long-term and importantly, they are almost entirely avoidable in the short-term.

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